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Taylor Series

Power Series

$$f(x) = c_0 + c_1 x + c_2 x^2 + \cdots \quad \text{——①}$$

$$= \sum_{n=0}^{\infty} c_n x^n$$

let $x = 0$,

$f(0) = c_0$

$f'(x) = 0 + c_1 + 2c_2 x + 3c_3 x^2 + \cdots$

$f'(0) = c_1 \qquad c_1 = \frac{f'(0)}{1} = \frac{f'(0)}{1!}$

$f''(x) = 2c_2 + 6c_3 x + \cdots$

$f''(0) = 2c_2 \qquad c_2 = \frac{f''(0)}{2} = \frac{f''(0)}{2!}$

$f'''(x) = 6c_3 + \cdots$

$f'''(0) = 6c_3 \qquad c_3 = \frac{f'''(0)}{3!}$

Hy, $c_n = \frac{f^n(x)}{n!}$

$$f(x) = f(0) + \frac{f'(0)}{1!}x + \frac{f''(0)}{2!}x^2 + \frac{f'''(0)}{3!}x^3 + \cdots$$

$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!} x^n$$ → MacLaurin Series

$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!} (x-a)^n$$ → Taylor Series

Some common Taylor series approximations

f(x) = sin(x) (centered at x=0)

	$f(x)$	$f'(x)$	$f''(x)$	$f'''(x)$	$f^{iv}(x)$	$f^{v}(0)$	$f^{vi}(0)$	$f^{vii}(0)$
	$\sin x$	$\cos x$	$-\sin x$	$-\cos x$	$\sin x$	$\cos x$	$-\sin x$	$-\cos x$
at $x=0$	0	1	0	-1	0	1	0	-1

$$f(x) = f(0) + \frac{f'(0)}{1!}x + \frac{f''(0)}{2!}x^2 + \cdots$$

$$= 0 + \frac{1}{1!}x + \frac{0}{2!}x^2 + \frac{-1}{3!}x^3 + \frac{0}{4!}x^4 + \frac{1}{5!}x^5 + \frac{0}{6!}x^6 - \frac{1}{7!}x^7 + \cdots$$

$$\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \frac{x^9}{9!} - \frac{x^{11}}{11!} + \cdots \quad \text{centred at } x=0$$

$$\sin x = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n+1}}{(2n+1)!}$$

About $x = \pi$:

$$f(x) = f(a) + \frac{f'(a)}{1!}(x-a) + \frac{f''(a)}{2!}(x-a)^2 + \cdots$$

$$f(x) = f(\pi) + \frac{f'(\pi)}{1!}(x-\pi) + \frac{f''(\pi)}{2!}(x-\pi)^2 + \cdots$$

	$f(x)$	$f'(x)$	$f''(x)$	$f'''(x)$	$f^{4}(x)$
	$\sin x$	$\cos x$	$-\sin x$	$-\cos x$	$\sin x$
at $x=\pi$	0	-1	0	1	0

$$f(x) = 0 + \frac{(-1)}{1!}(x-\pi) + \frac{0}{2!}(x-\pi)^2 + \frac{1}{3!}(x-\pi)^3 + \frac{0}{4!}(x-\pi)^4 + \cdots$$

$$f(x) = -(x-\pi) - \frac{(x-\pi)^3}{3!} - \cdots$$

f(x) = cos(x)

	$f(x)$	$f'(x)$	$f''(x)$	$f'''(x)$	$f^{4}(x)$	$f^{5}(x)$	$f^{6}(x)$	$f^{7}(x)$
	$\cos x$	$-\sin x$	$-\cos x$	$\sin x$	$\cos x$	$-\sin x$	$-\cos x$	$\sin x$
at $x=0$:	1	0	-1	0	1	0	-1	0

$$f(x) = f(0) + \frac{f'(0)}{1!}x + \frac{f''(0)}{2!}x^2 + \cdots$$

$$= 1 + 0 - \frac{1}{2!}x^2 + 0 + \frac{1}{4!}x^4 + 0 - \frac{1}{6!}x^6 + 0$$

$$= 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \frac{x^8}{8!} - \frac{x^{10}}{10!} + \cdots$$

$$\cos x = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n}}{(2n)!}$$

f(x) = exp(x)

	$f(x)$	$f'(x)$	$f''(x)$	$f'''(x)$	$f^{4}(x)$	$f^{5}(x)$	$f^{6}(x)$
	e^x	e^x	e^x	e^x	e^x	e^x	e^x
at $x=0$:	1	1	1	1	1	1	1

$$f(x) = f(0) + \frac{f'(0)}{1!}x + \frac{f''(0)}{2!}x^2 + \cdots$$

$$e^x = 1 + \frac{x}{1!} + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots$$

$$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!}$$

f(x) = log(1+x)

	$f(x)$	$f'(x)$	$f''(x)$	$f'''(x)$	$f^{4}(x)$	$f^{5}(x)$
	$\log(1+x)$	$\frac{1}{1+x}$	$\frac{-1}{(1+x)^2}$	$\frac{2}{(1+x)^3}$	$\frac{-6}{(1+x)^4}$	$\frac{24}{(1+x)^5}$
at $x=0$:	0	1	-1	2	-6	24

$$f(x) = 0 + \frac{1}{1!}x + \frac{-1}{2!}x^2 + \frac{2}{3!}x^3 + \frac{-6}{4!}x^4 + \frac{24}{5!}x^5 + \cdots$$

$$= x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \frac{x^5}{5} + \cdots$$

$$\log(1+x) = \sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n} x^n$$

Convergence of Taylor Series (Ratio Test)

Ratio test: $\lim_{n\to\infty}\left|\frac{u_{n+1}}{u_n}\right| = k$

if k < 1: convergent
k > 1: divergent
k = 1: ratio test fails to conclude

Q: $\sum_{n=0}^{\infty}\frac{x^n}{n!}$

→ $\frac{u_{n+1}}{u_n} = \frac{\frac{x^{n+1}}{(n+1)!}}{\frac{x^n}{n!}} = \frac{x}{(n+1)}$

$\lim_{n\to\infty}\left|\frac{u_{n+1}}{u_n}\right| = \lim_{n\to\infty}\left|\frac{x}{n+1}\right| = 0$

∴ It converges for all x.
Radius of convergence $= \infty$
Interval of convergence : $(-\infty, \infty)$

Q: $\sum_{n=0}^{\infty} n!\, x^n$

→ $\frac{u_{n+1}}{u_n} = \frac{(n+1)!\, x^{n+1}}{n!\, x^n} = (n+1)x$ $\quad \lim_{n\to\infty}|(n+1)x| = \infty$

converges only for $x = a$. $= 0$
radius of convergence $= 0$
Interval of convergence $= \{a\}$. $= \{0\}$

x^n is centred at 0.
So $a = 0$.

Q: $\sum_{n=0}^{\infty}\frac{(3x-2)^n}{n}$

→ centred at $x = 2/3$